

HOME TASK - ADVANCED SQL QUERIES

Aim :

To enhance SQL problem-solving abilities by solving practical database questions using joins, grouping, filtering, subqueries, and aggregate functions.

Objectives

- Understand practical SQL problem-solving techniques.
- Apply JOIN, GROUP BY, HAVING, and aggregate functions.
- Retrieve missing records using LEFT JOIN.
- Use filtering and sorting to generate analytical reports.
- Develop optimized SQL queries for real-world scenarios.

Question 1 (Easy):

Matching Skills Find candidates who possess all the required skills: Python, Tableau, and PostgreSQL.

Solution

```
SELECT candidate_id FROM candidates
WHERE skill IN ('Python','Tableau','PostgreSQL')
GROUP BY candidate_id HAVING COUNT(DISTINCT skill)=3
ORDER BY candidate_id;
```

Question 2 (Medium):

Pages With No Likes Find the IDs of Facebook pages that have never received a like.

Solution

```
SELECT p.page_id FROM pages p
LEFT JOIN page_likes pl
ON p.page_id=pl.page_id
WHERE pl.page_id IS NULL
ORDER BY p.page_id;
```

Question 3 (Hard):

Tesla Unfinished Parts Find the part names and the number of unfinished parts whose finish date is NULL.

Solution

```
SELECT part, COUNT(*) AS unfinished_parts
FROM parts_assembly
WHERE finish_date IS NULL
GROUP BY part
ORDER BY unfinished_parts DESC, part;
```

Learning Outcomes

- Write SQL queries using joins and aggregate functions.
- Apply GROUP BY and HAVING for grouped analysis.
- Retrieve unmatched records using LEFT JOIN.
- Solve real-world SQL interview problems efficiently.
- Improve database query writing and optimization skills.